

Exact optimization of the arithmetic full-jet determinant weights

13 September 2026

This is a separate continuation of the sealed arithmetic volume upper estimate AU.1–33 and AU.40–49. It solves the weight optimization in AU.19 completely, including boundary optima, zero polynomial coefficients, the sole nonunique case, and an evaluable one-dimensional characterization. The signed raw-jet minors, source coordinate phases, factorials and arithmetic masses remain explicit. The improvement over equal weights is proved to be less than one natural-logarithm unit for every degree within that specified family. The continuation (OW.30)–(OW.40) proves a stronger source estimate by keeping the even moments together. It then solves the entire resulting diagonal family using the actual moments through degree $2q$. Its exact comparison with AU.19 removes a term of order $q \log q$; the earlier less-than-one theorem retains its precise, narrower domain.

1 The original source and the signed full-jet minors

Keep $g = 2\xi$, the original complete monic zero packet h , and

$$w_h(t) = |(g/h)(1/2 + it)|^2/(2\pi), \quad m_k = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad S = c + iu, \quad c = k/2. \quad (\text{OW.1})$$

The source polynomial norm is $\int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du$ and its mass is μ_h^k . For the original degree- q monic relation χ put

$$\psi(u) = i^{-q} \chi(c + iu) = \sum_{j=0}^q \psi_j u^j, \quad \psi_q = 1, \quad \psi_j = \frac{i^{j-q}}{j!} \chi^{(j)}(c). \quad (\text{OW.2})$$

The last identity is the polynomial Taylor expansion at the original centre c . The exact ring map $P(S) \mapsto P(c + iu)$ has inverse $f(u) \mapsto f((S - c)/i)$ and intertwines multiplication by S with $c + iM_u$. Its raw derivative of order d at $\zeta = (\lambda - c)/i$ is $i^d P^{(d)}(\lambda)$. No coefficient phase or derivative factorial has been reassigned.

List the distinct roots ζ_a of ψ , in the original order, with their complete lengths ℓ_a , so $\sum_a \ell_a = q \geq 1$. For $N \geq q - 1$ the full raw-jet matrix is

$$(E_N)_{(a,d),j} = \begin{cases} j! \zeta_a^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d, \end{cases} \quad 0 \leq d < \ell_a, \quad 0 \leq j \leq N. \quad (\text{OW.3})$$

Taylor expansion proves that its kernel consists of polynomials divisible by ψ . Thus $Z := E_{q-1}$ is invertible by degree, and $E_q(\psi_0, \dots, \psi_q)^T = 0$. Write $\Delta_j = \det E_q[:, \{0, \dots, q\} \setminus \{j\}]$, with the retained increasing column order. The precise identity is

$$\Delta_j = (-1)^{q-j} \psi_j \det Z \quad (0 \leq j \leq q). \quad (\text{OW.4})$$

To prove the sign, append any row of E_q as a final row and expand the resulting zero determinant along that row. It shows that $((-1)^j \Delta_j)_{j=0}^q$ lies in the kernel of E_q . That kernel has dimension

one because its first q columns form Z . It is therefore a scalar multiple of the displayed monic coefficient vector. Its last coordinate is $(-1)^q \Delta_q = (-1)^q \det Z$, which proves (OW.4). In particular a zero coefficient gives exactly a zero minor, rather than a deleted column or shortened jet space.

The raw confluent determinant, in the same row order, is

$$\det Z = \left(\prod_a \prod_{d=0}^{\ell_a-1} d! \right) \prod_{a < a'} (\zeta_{a'} - \zeta_a)^{\ell_a \ell_{a'}}. \quad (\text{OW.5})$$

For completeness take the ordinary Vandermonde determinant with temporarily distinct nodes grouped into the prescribed clusters. Successive divided-difference row operations remove its within-cluster Vandermonde factors. The finite Taylor expansion of each monomial shows that its order- d divided-difference row tends to the raw derivative divided by $d!$. Multiplication of these rows by $d!$ gives (OW.5), with the cross-cluster factors still in their original order. Equivalently this proves the identity after cancellation of polynomial factors, so no analytic source approximation is involved.

Fix $0 < b < \pi/2$ and retain the two literal Laplace values

$$M_k = M_h(b)^k + M_h(-b)^k, \quad M_h(b) = \int_{\mathbb{R}} e^{bt} w_h(t) dt, \quad \beta_j = \frac{b^{2j}}{(2j)!} > 0. \quad (\text{OW.6})$$

For positive auxiliary weights τ_j the matrix used in AU.19 is $A_q(\tau) = E_q \text{diag}(\tau_j \beta_j) E_q^*$. Cauchy–Binet and (OW.4) prove

$$\begin{aligned} \det A_q(\tau) &= |\det Z|^2 \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \\ a_j &= |\psi_j|^2 \prod_{l \neq j} \beta_l = \left(\prod_{l=0}^q \beta_l \right) \eta_j, \quad \eta_j = \frac{|\psi_j|^2}{\beta_j} = \frac{(2j)!}{(j!)^2 b^{2j}} |\chi^{(j)}(c)|^2. \end{aligned} \quad (\text{OW.7})$$

In particular $a_q > 0$ and $\eta_q = (2q)! b^{-2q} > 0$. To verify Cauchy–Binet directly, expand the determinant multilinearly in the columns of E_q and then its adjoint. Repeated column choices vanish by alternation; collecting each q -element set gives its minor times its conjugate and the product of its diagonal weights. This yields exactly the first line of (OW.7). The signed identity is proved before taking the squared modulus.

For arbitrary positive τ , let $s = \sum_j \tau_j$ and $t_j = \tau_j/s$. The map $(\tau_j) \mapsto (s, (t_j))$ is a bijection from the positive cone to $(0, \infty)$ times the open simplex, with inverse $\tau_j = s t_j$. The determinant in (OW.7) acquires exactly the factor s^q . Thus the $q \log s$ in AU.19 cancels exactly. This changes auxiliary optimization coordinates and does not change m_k , its mass, or either source or quotient norm. Multiplication of all a_j by the common positive product in (OW.7) multiplies the objective by that product, with the maximizing weights unchanged. It suffices to solve the following finite optimization with all its original coefficients retained.

2 Global maximizers, including every zero coefficient

For $a_0, \dots, a_q \geq 0$ with $a_q > 0$, define

$$\mathcal{S}_q = \{\tau \in \mathbb{R}^{q+1} : \tau_j \geq 0, \sum_j \tau_j = 1\}, \quad D_a(\tau) = \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \quad d(a) = \max_{\tau \in \mathcal{S}_q} D_a(\tau). \quad (\text{OW.8})$$

The objective is a polynomial and the simplex is compact. Its value at equal weights is positive, so a positive maximum exists. Put $I = \{j : a_j > 0\}$, $p = |I|$, $A = \max_j a_j$, and $S_a = \sum_j a_j$.

Theorem 2.1 (Complete boundary and uniqueness classification). *For $q \geq 2$ the maximizing weights are precisely the following.*

1. *If $A \geq S_a - A$, choose its index j . Unless exactly two coefficients are positive and equal, the maximizer is unique: $\tau_j = 0$, $\tau_i = 1/q$ for every $i \neq j$, and $d(a) = A/q^q$.*
2. *If exactly two coefficients $a_j = a_l = A$ are positive, every maximizer, and only a maximizer, satisfies $\tau_j + \tau_l = 1/q$ and $\tau_i = 1/q$ at the other $q - 1$ indices. Its value is again A/q^q .*
3. *If $A < S_a - A$, then $p \geq 3$ and the maximizer is unique and has every coordinate strictly positive. At a zero coefficient it satisfies $\tau_j = 1/q$.*

For $q = 1$, $D_a = a_0\tau_1 + a_1\tau_0$ is linear: it is maximized at the endpoint selecting its larger coefficient, or everywhere if the two coefficients are equal. Its value is $\max(a_0, a_1)$.

Proof. An elementary determinant representation proves the needed concavity. Set $v_i = \sqrt{a_i/a_q}$ for $0 \leq i < q$ and

$$M(\tau) = \text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q vv^*. \quad D_a(\tau) = a_q \det M(\tau). \quad (\text{OW.9})$$

For positive first q weights, the rank-one determinant expansion gives $\det M = \prod_{i < q} \tau_i (1 + \tau_q \sum_{i < q} |v_i|^2 / \tau_i)$. This is (OW.9); since both sides are polynomials it holds at boundary weights too. Every $M(\tau)$ is positive semidefinite on the simplex, and $D_a > 0$ precisely when it is positive definite. For a real affine line $M(t) = M_0 + tE$ through positive definite matrices, differentiating the inverse identity gives $(M^{-1})' = -M^{-1}EM^{-1}$. The determinant's multilinear derivative therefore proves

$$(\log \det M)' = \text{Tr}(M^{-1}E), \quad (\log \det M)'' = -\text{Tr}((M^{-1/2}EM^{-1/2})^2) \leq 0. \quad (\text{OW.10})$$

The final matrix is Hermitian, so its squared eigenvalues are nonnegative, and equality holds exactly when $E = 0$.

If $p \geq 3$, at least two coordinates of v are nonzero. The linear map from τ to $M(\tau)$ is injective: an off-diagonal entry between two such coordinates forces a zero increment of τ_q whenever the matrix increment is zero; then its diagonal forces every other increment to be zero. Thus $\log D_a$ is strictly concave on its positive domain, which is convex as the inverse image of the positive definite cone. There can be at most one maximizer with positive objective.

If a simplex point has at least two zero coordinates, every monomial in D_a contains a zero, so its objective is zero. A positive boundary maximum must therefore have exactly one zero coordinate, say $\tau_j = 0$, and then $D_a = a_j \prod_{i \neq j} \tau_i$. It requires $a_j > 0$. For q nonnegative numbers of sum one, concavity of $\log x$ (whose second derivative is $-x^{-2}$) gives product at most q^{-q} , with equality exactly when they all equal $1/q$; a zero factor has product zero. Hence every positive boundary maximum is the candidate just described. At this point

$$\partial_i D_a = a_j / q^{q-1} \quad (i \neq j), \quad \partial_j D_a = (S_a - a_j) / q^{q-1}. \quad (\text{OW.11})$$

If $S_a - a_j > a_j$, the feasible path $\tau_j = \varepsilon$, $\tau_i = (1 - \varepsilon)/q$ for $i \neq j$ has positive derivative $(S_a - 2a_j)/q^{q-1}$ at zero. That boundary point cannot maximize. Conversely, if $S_a - a_j \leq a_j$, the gradient of $\log D_a$ at the candidate has all active entries equal and the inactive entry no larger. Its scalar product with any feasible displacement is nonpositive. Concavity along the segment to any point with positive objective gives $\log D_a(\tau) \leq \log D_a(\tau^*) + \nabla \log D_a(\tau^*) \cdot (\tau - \tau^*) \leq \log D_a(\tau^*)$. This derivative is valid at the boundary because $M(\tau^*) > 0$ and remains positive in a neighborhood. Points of zero objective cannot exceed it. This proves the boundary criterion and, for $p \geq 3$, uniqueness from strict concavity.

If $p = 1$, $D_a = A \prod_{i \neq j} \tau_i$ and the same product inequality gives the stated unique maximizer. If $p = 2$, with positive coefficients at j, l , factor the polynomial exactly as

$$D_a(\tau) = \left(\prod_{i \notin \{j, l\}} \tau_i \right) (a_j \tau_l + a_l \tau_j). \quad (\text{OW.12})$$

At fixed $\tau_j + \tau_l$, unequal coefficients allocate that sum entirely to the larger coefficient's complementary variable. The remaining q factors of total sum one have the unique equal product maximum. If the two coefficients coincide, their sum is one factor instead, and equality in that same product inequality gives exactly the listed segment of maximizers.

When $A < S_a - A$, no boundary candidate can maximize, so compactness gives an interior maximum and strict concavity makes it unique. It remains to verify the stated zero coordinate value; that follows from the complete interior stationarity calculation below. The case $q = 1$ is immediate from its displayed linear polynomial. $\square$

No artificial coefficient phase is required to apply (OW.9) to the original raw-jet matrix. In fact $E_q = Z[I_q, -\psi_{<q}]$ by monic remainder, so

$$A_q(\tau) = ZB_0^{1/2}(\text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q z z^*)B_0^{1/2}Z^*, \quad B_0 = \text{diag}(\beta_0, \dots, \beta_{q-1}), \quad z_i = -\psi_i \sqrt{\beta_q / \beta_i}. \quad (\text{OW.13})$$

Its $|z_i|^2$ equals η_i / η_q . Define the diagonal unitary $U_{ii} = -\psi_i / |\psi_i|$ when $\psi_i \neq 0$ and $U_{ii} = 1$ otherwise. Then $z = Uv$, with $v_i = \sqrt{\eta_i / \eta_q}$, and its central matrix is $UM(\tau)U^*$. This is the exact phase-preserving congruence connecting the auxiliary determinant proof to the original columns; all the phase entries remain in U .

3 A complete monotone scalar description of the interior optimum

Assume $q \geq 2$ and $A < S_a - A$, so the preceding theorem gives its unique interior maximizer. For positive weights put $P = \prod_j \tau_j$ and $S = \sum_j a_j / \tau_j$; then $D_a = PS$. Differentiating every factor gives $\partial_i D_a = P(S / \tau_i - a_i / \tau_i^2)$. The Lagrange multiplier for $\sum \tau_i = 1$ is qD_a because $\sum_i \tau_i \partial_i D_a = qD_a$, each monomial having degree q . Consequently its complete equations are

$$a_i = S \tau_i (1 - q \tau_i). \quad (\text{OW.14})$$

If $a_i = 0$, positivity gives $\tau_i = 1/q$. At a positive coefficient let $y_i = 1 - q \tau_i$. The equation shows $0 < y_i < 1$, and counting the zero coordinates gives

$$\sum_{i \in I} y_i = 1, \quad a_i = c_* y_i (1 - y_i), \quad c_* = S/q > 0, \quad \tau_i = (1 - y_i)/q. \quad (\text{OW.15})$$

Here c_* is the optimizer's scalar, distinct from the source centre $c = k/2$ in (OW.1). Conversely any positive solution of (OW.15), together with $\tau_i = 1/q$ at zero coefficients, has sum one and satisfies (OW.14): indeed $\sum_i a_i / \tau_i = qc_* \sum_{i \in I} y_i = qc_*$. Concavity then proves that this stationary point is the global maximizer, and uniqueness has already been established.

For $0 \leq t \leq 1/A$ and $i \in I$ define the continuous functions

$$f_i(t) = \frac{1 - \sqrt{1 - a_i t}}{2} = \frac{a_i t}{2(1 + \sqrt{1 - a_i t})}, \quad F(t) = \sum_{i \in I} f_i(t), \quad t = 4/c_*. \quad (\text{OW.16})$$

All square roots are the nonnegative real roots. The second formula is its exact rationalization, valid also at $t = 0$. Each f_i is strictly increasing on this interval, has value zero at zero, and takes values at most $1/2$.

Theorem 3.1 (The unique scalar root and the complete maximizing value). *Under $A < S_a - A$ exactly one of the following descriptions applies.*

1. *If $F(1/A) \geq 1$, there is a unique $t_* > 0$ in $(0, 1/A]$ with $F(t_*) = 1$. Set $y_i = f_i(t_*)$ at every positive coefficient.*
2. *If $F(1/A) < 1$, the largest coefficient has a unique index j . There is a unique $t_* \in (0, 1/A)$ solving*

$$H(t) := \sum_{i \in I \setminus \{j\}} \frac{a_i}{A} \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = 1. \quad (\text{OW.17})$$

The function H is strictly decreasing, with $H(0) = (S_a - A)/A > 1$ and $H(1/A) = 2F(1/A) - 1 < 1$. Set $y_j = 1 - f_j(t_)$ and $y_i = f_i(t_*)$ for $i \neq j$.*

In either case the weights are those in (OW.15), with zeros handled as stated, and the exact value is

$$d(a) = \frac{4}{q^q t_*} \prod_{i \in I} (1 - y_i). \quad (\text{OW.18})$$

Proof. In (OW.15) at most one y_i can exceed $1/2$ because their sum is one. If none does, its quadratic equations have precisely the smaller roots $f_i(t)$, and their sum is $F(t) = 1$. Continuity, strict increase and $F(0) = 0$ prove existence and uniqueness exactly in the first listed case.

If $y_j > 1/2$, each other y_i is strictly smaller than $1 - y_j < 1/2$, because $p \geq 3$ gives at least two other positive terms. The function $x(1 - x)$ is strictly increasing on $(0, 1/2)$. It follows that $a_i < c_* y_j (1 - y_j) = a_j$, so the large root must occur at the unique largest coefficient. Its sum equation is $\sum_{i \neq j} f_i(t) = f_j(t)$. At $t > 0$ dividing by the positive f_j and using the second formula in (OW.16) gives exactly (OW.17).

If $F(1/A) < 1$, the maximum is unique already from the endpoint values: two largest coefficients would contribute $1/2 + 1/2$ and the other positive terms would make $F(1/A) > 1$. For $a_i < A$ and $0 < t < 1/A$, direct logarithmic differentiation gives

$$\frac{d}{dt} \log \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = \frac{(1 - a_i t)^{-1/2} - (1 - At)^{-1/2}}{2t} < 0. \quad (\text{OW.19})$$

Thus every term of H decreases strictly. Its value at zero is given by continuity, and $f_j(1/A) = 1/2$ gives the second endpoint value. The intermediate value theorem proves its unique root. The constructed y has positive entries summing to one and solves the quadratic equations, so (OW.15) proves it is the optimum. Conversely if $F(1/A) \geq 1$, an index with a large root would have to be the unique largest one; its decreasing H would have $H(1/A) = 2F(1/A) - 1 \geq 1$ and hence no root inside $(0, 1/A)$. At equality, the largest root is exactly $1/2$ and coincides with the first construction. This proves completeness of the cases.

Finally $\prod_j \tau_j = q^{-(q+1)} \prod_{i \in I} (1 - y_i)$ and $S = qc_* = 4q/t_*$. Multiplication gives (OW.18), with the full original dimension exponent q retained. $\square$

The zero-coefficient assertion can also be proved globally without interior differentiation. If $p \geq 2$ and $z = q + 1 - p > 0$, write $s = \sum_{i \in I} \tau_i$ and $\tau_i = s\sigma_i$ on I . The polynomial factors as $(\prod_{i \notin I} \tau_i) s^{p-1} D_{a|I}(\sigma)$. For fixed s , the first product is at most $((1 - s)/z)^z$. The derivative of $\log((1 - s)^z s^{p-1})$ vanishes only at $s = (p - 1)/q$, and its second derivative is strictly negative. The endpoints have zero product. Hence every maximum has $\tau_i = 1/q$ at zero coefficients, and

$$d(a) = \frac{(p - 1)^{p-1}}{q^q} d_{p-1}(a|I), \quad (\text{OW.20})$$

where the last maximum uses the explicitly displayed positive-index simplex of dimension $p - 1$. If $z = 0$, the same formula has factor one. For $p = 1$, the first theorem gives the value directly,

with the positive coordinate zero and the other q weights $1/q$. Thus the reduction has an explicit map and determinant power, without changing the dimension of the original arithmetic quotient.

For three positive coefficients a, b, d satisfying the strict triangle inequalities, the solution even has a rational expression. Put $\Delta = 2ab + 2ad + 2bd - a^2 - b^2 - d^2 > 0$. Then

$$\tau_a = \frac{2a(b+d-a)}{q\Delta}, \quad \tau_b = \frac{2b(a+d-b)}{q\Delta}, \quad \tau_d = \frac{2d(a+b-d)}{q\Delta},$$

$$\tau_i = 1/q \quad (a_i = 0), \quad \boxed{d(a) = \frac{4abd}{q^q\Delta}}. \quad (\text{OW.21})$$

Indeed $\Delta = 4ab - (a+b-d)^2 > 0$ because $0 < a+b-d < 2\min(a, b) \leq 2\sqrt{ab}$. On the positive-index simplex the proposed coordinates $\sigma_a = a(b+d-a)/\Delta$ and its cyclic counterparts are positive and sum to one. For its quadratic polynomial $a\sigma_b\sigma_d + b\sigma_a\sigma_d + d\sigma_a\sigma_b$, direct substitution makes every gradient coordinate $2abd/\Delta$. Euler's degree-two identity then gives its value abd/Δ ; concavity proves maximality. Equation (OW.20) supplies exactly the powers and the weights in (OW.21).

4 Evaluable enclosures and the exact maximum gain

For exact rational nonnegative input coefficients, the branch comparison at $1/A$ uses only finitely many nonnegative square roots. Equality is permitted as the first branch's endpoint. In the strict cases, (OW.16) or (OW.17) has a continuous strictly monotone scalar function on a known compact interval with opposite endpoint signs. Rational bisection therefore gives its unique root. Square roots themselves admit rational brackets by squaring positive rationals and bisecting. If an exact trial point is a root, it already solves the problem. Otherwise convergent radical brackets eventually resolve the nonzero sign at that trial point. A positive lower bracket for t_* is obtained after finitely many bisections because $t_* > 0$. Substitution in (OW.18) then yields convergent value brackets; every denominator is bounded away from zero on those root brackets. Boundary maximizers require only the exact coefficient comparison $A \geq S_a - A$ and their explicit product formula.

There is also a uniform certificate that avoids deciding a branch from uncertain coefficients. For two nonnegative coefficient lists, the product inequality used in (OW.11) gives, at every simplex point,

$$|D_a(\tau) - D_{\hat{a}}(\tau)| \leq q^{-q} \sum_j |a_j - \hat{a}_j|.$$

Taking a maximizing point for each list in turn proves

$$\boxed{|d(a) - d(\hat{a})| \leq q^{-q} \sum_j |a_j - \hat{a}_j|}. \quad (\text{OW.22})$$

The maximum is also increasing in each coefficient because each monomial is nonnegative. Thus nonnegative rational boxes $a^- \leq a \leq a^+$ give $d(a^-) \leq d(a) \leq d(a^+)$, and their value width is at most $q^{-q} \sum (a_j^+ - a_j^-)$. These statements remain valid if a corner list is all zero, where its maximum is exactly zero, or its reference last coefficient vanishes: a positive reference coefficient can be selected by the explicit simultaneous permutation of coefficient and simplex indices, whose inverse permutation restores every original index. No unresolved arithmetic equality must be assumed to construct the enclosure. For the original list $a_q > 0$ its lower value eventually stays positive, so logarithms can also be enclosed by their ordinary monotone integral $\log x = \int_1^x dt/t$.

A rational candidate has a direct exact value certificate, so no radical equality needs to be decided merely to enclose the answer. At any simplex point τ with $D := D_a(\tau) > 0$, calculate the full polynomial gradient $g_i = \partial_i D_a(\tau)$ and set

$$h = \max_i g_i - qD \geq 0.$$

The sign follows from $\sum_i \tau_i g_i = qD$. Concavity and the simplex constraint give, for every point of positive objective,

$$\log D_a(\sigma) \leq \log D + \frac{\sum_i g_i(\sigma_i - \tau_i)}{D} \leq \log D + h/D.$$

Therefore

$$\boxed{D \leq d(a) \leq De^{h/D}, \quad h < D \implies d(a) \leq \frac{D^2}{D-h}.} \quad (\text{OW.22a})$$

The last inequality follows from $e^x = \sum_{n \geq 0} x^n/n! \leq \sum_{n \geq 0} x^n = (1-x)^{-1}$ for $0 \leq x < 1$. The polynomial gradient is defined also at a boundary point; the argument is valid there because its matrix (OW.9) is positive definite. Thus an exactly stationary rational candidate, including a boundary maximum with the correct inactive gradient, has $h = 0$ and certifies its exact value.

Here is a terminating way to combine this certificate with scalar evaluation, including an unresolved equality $F(1/A) = 1$. Use rational brackets for every square root. If the endpoint interval separates from one, select the proved branch and bisect its scalar root. If an endpoint interval still contains one, also form the candidate from its smaller-root values at $1/A$. At any scalar trial whose comparison interval contains zero, form its candidate as well, instead of making an unsupported sign decision. Rational midpoints of the root brackets give nonnegative y approximations; replace the corresponding nonnegative auxiliary weights by their ratios to their positive sum, making an exact rational simplex point. One may keep zero-coefficient weights exactly $1/q$ and apply the positive-index rescaling (OW.20) only to the remaining coordinates. These are the explicit positive-cone coordinate maps already proved, and no source measure changes. Test (OW.22a) at each such candidate.

As square-root intervals narrow, a strictly nonzero branch or bisection sign is eventually determined. If a tested equality is exact, its candidate tends instead to the already proved optimum. In a fixed strict branch, successive root brackets tend to its unique root. In all cases the resulting candidates tend to the maximizer or to a member of the stated maximizing segment. There $D > 0$ and $h = 0$ by (OW.11) or (OW.14), and D, h are continuous functions of the candidate. Thus eventually $h < D$ and the rational certificate width $D^2/(D-h) - D$ is smaller than any prescribed positive rational tolerance. Every returned interval is certified solely by the rational polynomial gradient and (OW.22a), independently of the numerical branch-selection history.

Theorem 4.1 (The finite gain is uniformly less than one). *Let $\tau_j^{\text{eq}} = 1/(q+1)$ and $S_a = \sum a_j > 0$. For every $q \geq 1$,*

$$D_a(\tau^{\text{eq}}) = \frac{S_a}{(q+1)^q}, \quad \frac{S_a}{(q+1)^q} \leq d(a) \leq \frac{S_a}{q^q},$$

$$\boxed{0 \leq \mathfrak{g}(a) := \log \frac{d(a)}{D_a(\tau^{\text{eq}})} \leq q \log(1 + 1/q) < 1.} \quad (\text{OW.23})$$

The lower equality holds precisely when all coefficients are equal. The upper non-strict equality holds precisely when exactly one coefficient is positive.

Proof. The equal-weight value is obtained by substituting each weight. Every product of q selected nonnegative weights has sum at most one and is therefore at most q^{-q} . Summing with coefficients a_j gives the upper bound. If just one coefficient is positive, its boundary maximizer attains that bound. If two or more are positive, equality would require every corresponding product to attain q^{-q} . For the term omitting j , that requires $\tau_j = 0$ and $\tau_i = 1/q$ at every other index. Two distinct positive terms cannot meet those two incompatible conditions. Thus the upper equality has exactly the stated case.

If equal weights maximize, they are an interior point. Differentiating the polynomial there gives $\partial_i D_a = (q+1)^{-(q-1)}(S_a - a_i)$, which must be constant in i under the sum constraint. Thus

every a_i is equal. Conversely equal coefficients give equal gradient coordinates at equal weights, and the determinant concavity proves maximality for $q \geq 2$; for $q = 1$ it is the constant linear polynomial. Finally $\log(1+x) < x$ for $x > 0$, since $\log(1+x) = \int_0^x (1+t)^{-1} dt < \int_0^x dt = x$. Apply this with $x = 1/q$ to prove the strict upper bound one. $\square$

One additional fully explicit lower bracket for the gain is

$$\mathbf{g}(a) \geq \max\{0, q \log(1 + 1/q) + \log(A/S_a)\}. \quad (\text{OW.24})$$

It follows by testing the best boundary point. In the boundary regime its second expression is the exact gain. When $p \geq 2$, testing equal weights on the positive-index simplex in (OW.20) also gives $\mathbf{g}(a) \geq q \log(1 + 1/q) + (p-1) \log((p-1)/p)$, with every zero retained through the factor in (OW.20).

5 The optimized original arithmetic bound, including boundary weights

To specify the analytic comparator used in the original bound, retain the proved constants $c_h > 0$, $\vartheta_h > 0$, $B = 42 + 2 \deg h$, $\alpha = \pi/2$, and $k \geq 3$. Its measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du \leq m_k(u) du. \quad (\text{OW.25})$$

Let H_N^ν and H_N be the two monomial source Grams and $K_N^{\nu, \text{raw}} = E_N(H_N^\nu)^{-1} E_N^*$, with the original E_N and ψ . Their positive definiteness follows from their positive almost-everywhere densities and the finite zero set of every nonzero polynomial. Contraction of the identity polynomial map gives $H_N^\nu \preceq H_N$, so $K_N^{\text{raw}} \preceq K_N^{\nu, \text{raw}}$. If J_N is monic remainder, the original quotient metric is $G_N = (J_N H_N^{-1} J_N^*)^{-1}$, $V_N = \det G_N$. Every least lift is $H_N^{-1} J_N^* G_N x$; its difference from any other lift is orthogonal to it, as multiplication by J_N verifies. Thus these are the original quotient norms. Since $E_N = Z J_N$, $\det K_N^{\text{raw}} = |\det Z|^2 / V_N$, exactly as in the signed jet map above. The comparison measure defines V_N^ν by the same formula, with its own literal mass.

The exponential-series inequality and convolution give $\int |u|^{2j} m_k(u) du \leq M_k / \beta_j$. Weighted Cauchy–Schwarz, with positive weights of sum one, then gives $H_q \preceq M_k \text{diag}(1/(\tau_j \beta_j))$. Inversion and the original raw-jet map prove $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau)$. Together with the upper comparison at degree $2q-1$ this proves the original AU.19 bound on the open simplex:

$$C_k := \log(V_q/V_{2q-1}) \leq q \log M_k + \log \det K_{2q-1}^{\nu, \text{raw}} - \log \det A_q(\tau). \quad (\text{OW.26})$$

At a boundary maximizer in the first theorem its omitted coefficient is positive, so (OW.7) shows $\det A_q(\tau^*) > 0$. Use the explicit positive regularization $\tau^\varepsilon = (1-\varepsilon)\tau^* + \varepsilon\tau^{\text{eq}}$ for $0 < \varepsilon < 1$. The raw matrices converge entry by entry to the positive definite $A_q(\tau^*)$. Taking the limit of their quadratic-form inequality preserves $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau^*)$. Continuity of the positive determinant therefore extends (OW.26) to the boundary optimum. No expression $1/\tau_j$ is evaluated at zero; the boundary statement is proved on the dual raw kernel.

Let $B_\beta = \prod_{j=0}^q \beta_j$. Substitution of the exact maximum (OW.7), followed by the already proved raw-jet determinant conversion, gives the sharpened original bound

$$C_k \leq \mathcal{U}_{\text{opt}} := q \log M_k - \log B_\beta - \log d(\eta) - \log V_{2q-1}^\nu. \quad (\text{OW.27})$$

Here $d(\eta)$ is completely evaluated by the first theorem and (OW.16)–(OW.18), including every coefficient zero and the actual factorials in (OW.7). The factor $|\det Z|^2$ cancels between the two displayed raw kernels only after its signed minors and phase map have been proved. The original source mass stays in M_k , and the comparison mass stays in V_{2q-1}^ν . If desired the latter determinant is computed in its exact source/relation form $V_{2q-1}^\nu = \det H_{2q-1}^\nu / \det(B_{2q-1}^* H_{2q-1}^\nu B_{2q-1})$, where

B_{2q-1} is multiplication by the same monic ψ . The quotient-first monomial-plus-relation coefficient matrix has determinant one; eliminating its relation block proves this formula.

The equal-weight upper expression from the identical estimate is

$$\mathcal{U}_{\text{eq}} = q \log M_k - \log B_\beta - \log \left(\frac{\sum_j \eta_j}{(q+1)^q} \right) - \log V_{2q-1}^\nu.$$

Thus its exact improvement is

$$\boxed{\mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} = \mathfrak{g}(\eta), \quad 0 \leq \mathfrak{g}(\eta) \leq q \log(1 + 1/q) < 1.} \quad (\text{OW.28})$$

This is an absolute bound, not a bound per coordinate or per degree. For the original complete quartet, $q = [1 + k(m-1)](k+1)^2$ tends to infinity. Dividing (OW.28) by $q \log k$ or by q^2 proves that the difference tends to zero on either scale. This conclusion requires no bound at all on the unknown comparison determinant's own asymptotic size.

For that quartet the gain is strictly positive and strictly smaller than the stated bound. Indeed its sum roots in the u chart are $(2b-k)\gamma - i(2a-k)\delta$, $0 \leq a, b \leq k$, each with the full length $1 + k(m-1)$. They are stable under negation, so comparing leading coefficients gives $\psi(-u) = (-1)^q \psi(u)$. Hence coefficients of the opposite parity vanish, while $\psi_q = 1$ is nonzero. The coefficients η_j cannot all be equal. Also at least one root is nonzero because $\delta, \gamma > 0$, so ψ is not the monomial u^q and at least two coefficients are nonzero. Applying both equality classifications in (OW.23) proves

$$\boxed{0 < \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} < q \log(1 + 1/q) < 1} \quad \text{for every admitted complete quartet.} \quad (\text{OW.29})$$

The nonmonomial assertion follows directly from the factorizations: the roots of u^q are all zero, whereas the displayed root list contains a nonzero root. Its exact replacement here is the signed cofactor map (OW.4), whose nonzero extra coefficient yields the additional positive determinant term. No relation or jet is removed.

6 Joint even moments and exact optimization of their diagonal family

Keep the source, chart, polynomial and every raw-jet row from (OW.1)–(OW.7). For $0 \leq j \leq q$ write

$$\mu_{2j} := \int_{\mathbb{R}} u^{2j} m_k(u) du > 0, \quad \mu_0 = \mu_h^k, \quad S_q(b) := \sum_{j=0}^q \beta_j \mu_{2j}. \quad (\text{OW.30})$$

The inequalities are strict because the source density is positive almost everywhere and $u^{2j} > 0$ off zero. Finiteness follows from (OW.6) and its exponential-series majorant. These are moments in the original u chart; the zeroth moment remains the full k -fold source mass.

For a coefficient vector $p = (p_0, \dots, p_q)^T$ the ordinary complex Cauchy–Schwarz inequality gives the pointwise statement

$$\left| \sum_{j=0}^q p_j u^j \right|^2 \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \left(\sum_{j=0}^q \beta_j u^{2j} \right) \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \cosh(bu).$$

The first inequality uses the two vectors $p_j/\sqrt{\beta_j}$ and $\sqrt{\beta_j}u^j$, with complex conjugation in the inner product; the second uses the full nonnegative even exponential series. Its finite tail is strictly positive for every $u \neq 0$. Integration, including both original Laplace values, therefore proves

$$\begin{aligned} H_q &\preceq S_q(b) \text{diag}(\beta_j^{-1}) \prec \frac{M_k}{2} \text{diag}(\beta_j^{-1}), \\ 0 < S_q(b) &< \int_{\mathbb{R}} \cosh(bu) m_k(u) du = \frac{M_h(b)^k + M_h(-b)^k}{2} = \frac{M_k}{2}. \end{aligned} \quad (\text{OW.31})$$

The second matrix inequality is strict positive definiteness because it is a strictly positive scalar difference times a positive diagonal matrix. The convolution identity here follows by integrating $e^{b(t_1+\dots+t_k)}$ against $\prod_i w_h(t_i)dt_i$; its nonnegative integrand permits iterated integration. The identical calculation applies at $-b$.

Inverting (OW.31) and applying the unchanged full raw-jet matrix gives

$$K_q^{\text{raw}} \succeq S_q(b)^{-1} E_q \text{diag}(\beta_j) E_q^*.$$

The determinant identity (OW.7) at the literal weights all equal to one is $|\det Z|^2 B_\beta \sum_j \eta_j$. The original numerator comparison $K_{2q-1}^{\text{raw}} \preceq K_{2q-1}^{\nu, \text{raw}}$ is unchanged. Hence

$$\begin{aligned} C_k &\leq \mathcal{U}_{\text{fin}} := q \log S_q(b) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu, \\ \mathcal{U}_{\text{fin}} &< \mathcal{U}_{\text{cosh}} := q \log(M_k/2) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu. \end{aligned} \quad (\text{OW.32})$$

Thus the pointwise finite sum supplies its own proved source bound; it does not use the older scalar estimate $M_k \sum \tau_j$ from AU.19. Direct subtraction, with no moment or factorial discarded, gives

$$\begin{aligned} \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{cosh}} &= q \log(2(q+1)), \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{fin}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} > 0. \end{aligned} \quad (\text{OW.33})$$

The positivity follows from $\mathfrak{g}(\eta) < 1$ and $q \log(2(q+1)) \geq \log 4 > 1$ for $q \geq 1$. For the last elementary inequality, $\log 4 = 2 \int_1^2 dt/t > 2(2-1)/2 = 1$.

The same argument determines a larger family completely. To avoid identifying these new diagonal weights with the cofactor coefficients a_j in (OW.7), denote them by $r_j > 0$ and put

$$s(r) := \sum_{j=0}^q r_j \mu_{2j}, \quad H_q \preceq s(r) \text{diag}(r_j^{-1}), \quad K_q^{\text{raw}} \succeq s(r)^{-1} E_q \text{diag}(r_j) E_q^*. \quad (\text{OW.34})$$

Indeed pointwise Cauchy–Schwarz with $p_j/\sqrt{r_j}$ and $\sqrt{r_j}u^j$, followed by integration, gives the middle inequality for every coefficient vector p . Inversion reverses the positive definite matrix order: if $X \preceq Y$, then $Y^{-1/2}XY^{-1/2} \preceq I$ and its positive eigenvalues are at most one, so $X^{-1} \succeq Y^{-1}$. Congruence by E_q proves the final inequality. This explicitly specifies the source-to-raw-jet map.

The positive-cone map and its inverse are

$$(r_j) \mapsto \left(s(r), t_j = \frac{r_j \mu_{2j}}{s(r)} \right), \quad (s, t) \mapsto r_j = \frac{s t_j}{\mu_{2j}}. \quad (\text{OW.35})$$

They are mutually inverse bijections between $\mathbb{R}_{>0}^{q+1}$ and $(0, \infty) \times \mathcal{S}_q^\circ$. Positivity of every moment proves each division is defined; summing the displayed t_j proves the simplex constraint, and substituting either composite restores every original index. This map acts only on auxiliary diagonal weights. It does not divide the source measure by its mass or alter a polynomial, jet, source coordinate, or quotient coordinate.

Put $\kappa_j = |\psi_j|^2 \mu_{2j}$, so $\kappa_q = \mu_{2q} > 0$. Applying (OW.4) and Cauchy–Binet before any phase cancellation gives

$$\begin{aligned} \frac{\det(E_q \text{diag}(r_j) E_q^*)}{s(r)^q} &= |\det Z|^2 \sum_{j=0}^q |\psi_j|^2 \prod_{l \neq j} \frac{r_l}{s(r)} \\ &= |\det Z|^2 \left(\prod_{j=0}^q \mu_{2j}^{-1} \right) D_\kappa(t). \end{aligned} \quad (\text{OW.36})$$

The product has exactly $q + 1$ moments. In each degree- q monomial its missing moment is supplied by the factor μ_{2j} in κ_j ; thus it includes, without a changed exponent, $\mu_0 = \mu_h^k$. Every original coefficient phase still occurs in (OW.4) and (OW.13), and each zero ψ_j gives its precise zero coefficient κ_j .

Combining (OW.34)–(OW.36) with the numerator comparison and the exact raw-jet determinant conversion gives the complete optimized bound

$$C_k \leq \mathcal{U}_{\text{mom}} := \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log V_{2q-1}^\nu. \quad (\text{OW.37})$$

The earlier global optimizer theorems apply to the literal list $\kappa_0, \dots, \kappa_q$, so (OW.8)–(OW.22a) calculate $d(\kappa)$ with all its boundary cases, scalar root, moment dependence and raw polynomial multiplicities retained. They also prove that (OW.37) is the infimum over the entire positive diagonal family (OW.34). For completeness at a boundary optimum t^* the limiting raw matrix is $E_q \text{diag}(t_j^*/\mu_{2j}) E_q^*$, whose determinant is positive by (OW.36). Regularize t^* by $(1 - \varepsilon)t^* + \varepsilon t^{\text{eq}}$ and use $r_j = t_j/\mu_{2j}$, so $s(r) = 1$. For every $\varepsilon > 0$ (OW.34) holds, and the raw matrices converge entrywise to that positive definite limit. Their quadratic-form inequalities and determinants pass to the limit exactly as in (OW.26). No inverse of a zero boundary weight is evaluated.

The finite even-series weights $r_j = \beta_j$ correspond under (OW.35) to the explicit interior point

$$t_j^\beta = \frac{\beta_j \mu_{2j}}{S_q(b)}, \quad D_\kappa(t^\beta) = \left(\prod_{j=0}^q \mu_{2j} \right) \frac{B_\beta \sum_j \eta_j}{S_q(b)^q}. \quad (\text{OW.38})$$

This equality follows term by term from the defining polynomial; it also follows by setting $r = \beta$ in (OW.36). Since its value is positive, all the following logarithms are defined. Their exact comparison is

$$\begin{aligned} \mathcal{U}_{\text{fin}} - \mathcal{U}_{\text{mom}} &= \log \frac{d(\kappa)}{D_\kappa(t^\beta)} \geq 0, \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{mom}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} + \log \frac{d(\kappa)}{D_\kappa(t^\beta)}. \end{aligned} \quad (\text{OW.39})$$

The last logarithm is exactly computable from the proved optimizer; no less-than-one claim is made for it relative to the generally nonuniform point t^β . If the moment simplex instead uses its own equal point t^{eq} , its gain is precisely $\mathfrak{g}(\kappa) < 1$ by (OW.23). These different reference points are related by the explicit map (OW.35), rather than silently identified.

The exact removed term $q \log(2(q+1))$ has order $q \log q$. For the original $q = [1 + k(m-1)](k+1)^2$ and fixed multiplicity m , its ratio to $q \log k$ tends to 2 if $m = 1$ and to 3 if $m > 1$. Indeed $q = (k+1)^2$ in the first case; in the second $q/k^3 = (m-1 + 1/k)(1 + 1/k)^2 \rightarrow m-1 > 0$. Taking logarithms proves both limits, while $\log(2(q+1))/q \rightarrow 0$ proves its ratio to q^2 tends to zero. The literal comparison (OW.39) retains all additional finite moment gains. It does not determine their growth or the comparison determinant's growth. In particular it does not subtract a term from AU's separately derived interval scalar bound without a proved comparison to that bound. It gives a stronger, fully specified source inequality and its original quotient-volume consequence using the actual finite moments.

There is also an exact finite comparison to the full original low-degree kernel, retaining all mixed moments. Let $\psi = (\psi_0, \dots, \psi_q)^T$ and let the coefficient matrix T have columns $1, u, \dots, u^{q-1}, \psi$ in the monomial basis of degree at most q . Its upper block is the identity and its last diagonal entry is $\psi_q = 1$, so $\det T = 1$. The last vector spans the relation kernel of J_q . In the Gram $T^* H_q T$, eliminate that last block by its positive scalar $\psi^* H_q \psi$. The resulting Schur complement is the Gram of the least lifts of the first q quotient coordinates: subtracting the orthogonal projection onto the relation is exactly the minimization defining G_q . Taking

determinants therefore proves

$$V_q = \frac{\det H_q}{\psi^* H_q \psi}, \quad \mathcal{U}_{\text{low}} := \log V_q - \log V_{2q-1}^\nu, \quad (OW.40)$$

$$0 \leq \mathcal{U}_{\text{mom}} - \mathcal{U}_{\text{low}} = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log \det H_q + \log(\psi^* H_q \psi).$$

The sign follows directly from the limiting raw-kernel inequality used in (OW.37): its determinant gives $|\det Z|^2/V_q \geq |\det Z|^2(\prod_j \mu_{2j}^{-1})d(\kappa)$. Its cancellation is legitimate because Z is invertible by (OW.3). The numerator comparison also gives $C_k \leq \mathcal{U}_{\text{low}}$. Thus (OW.40) records the exact nonnegative finite loss of the fully optimized diagonal source estimate relative to the full original H_q kernel. All entries of H_q are actual moments through degree $2q$ and every determinant is on its declared original space; the formula imposes no estimate on the loss's asymptotic growth.

7 Verification boundary and continuity

The sealed AU article is an input and remains unmodified. All new mathematics is written in (OW.1)–(OW.40): a signed full-jet identity, global maximum classification, monotone scalar evaluation, complete uncertain-input enclosures, the exact gain within AU.19, and the stronger joint-moment source family with its complete optimum. The auxiliary weight simplex and all its coordinate maps are specified separately from the original arithmetic source; its supported relations, Taylor unit, and quotient action are unchanged because every matrix uses the same E_N, J_N, ψ and (OW.2). A quotient-killed class remains the supported zero at its original label, and the induced fibre map fixes the external absence τ ; the optimization variables τ_j in this paper are real auxiliary weights with their own declared domain (OW.8).

Finite fixtures accompany the written proof and test its exact maps and optimum formulas on declared coefficient lists. They do not infer an arithmetic asymptotic from numerical optimization, assert new zero data, or replace the complete global proofs above.